FloatingDust FX (URP) - User Manual & Setup Guide

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1. Introduction & Requirements

Thank you for downloading FloatingDust FX! This package provides a lightweight, optimized atmospheric dust system designed specifically for the Universal Render Pipeline (URP). Featuring real-time spherical volume spawning, organic turbulence drift, and a built-in camera distance fade system, this effect instantly brings life, depth, and rich atmosphere to corridors, rooms, and open environments.

Built with industry-standard practices, our dual-layer particle architecture ensures a dense, natural distribution of dust motes that float freely in World Space while dynamically fading out as the camera approaches, preventing visual clutter and keeping your gameplay view clean.

Technical Requirements

- Minimum Unity Version: Unity 2022.3 LTS (or higher).
- Render Pipeline: Universal Render Pipeline (URP).
- Required Package: Visual Effect Graph (com.unity.visualeffectgraph) must be installed in your project via the Package Manager.

2. Package Contents & Capacity Limit

Everything is cleanly organized and compartmentalized inside the GDurell/FloatingDustFX/ hierarchy. You can safely delete the Demo folder when you are ready to build your final game.
- Documentation/: Contains this user manual.

- Effects/: Contains the main Visual Effect Asset (VFX_FloatingDust).
- Textures/: Particle textures.
- Demo/: A ready-to-play showcase scene (FloatingDust_ShowcaseScene) featuring an ambient setup to preview the volume.

Note on Dual-Layer System & Particle Capacity Limit

To achieve a rich, cinematic look without performance overhead, the system utilizes an internal dual-layer particle composition:
1. Micro-Motes (High Density): Tiny, subtle points with high baseline visibility that create the main atmospheric density.
2. Volumetric Specks (Low Density): Larger, softer particles with higher opacity variations interspersed throughout the volume to provide physical depth and scale.

The Visual Effect Graph particle pool is capped at a maximum capacity of 2,000 particles, allowing you to scale density for large rooms without affecting performance.

3. Quick Setup & How It Works

Because the effect is driven entirely by Unity's Visual Effect Graph, setup takes less than a minute.
1. Verify Package Installation: Ensure that the Visual Effect Graph package is installed in your project via the Unity Package Manager (Window > Package Manager).
2. Place in Scene: Simply drag the VFX_FloatingDust asset or prefab directly into any room, corridor, or environment location where you want atmospheric dust floating.
3. Automatic Camera Proximity Fade: The particle system includes a built-in distance fade and camera kill sphere. As the camera moves through or gets close to the volume, individual particles cleanly dissolve to zero alpha, preventing unpleasant clipping or view obstruction.

4. How to Use & Scene Integration

We designed this asset for maximum developer ergonomics.
- Zero Scripting Required: You do not need to pass position vectors or write custom scripts. Place the volume in your scene and let the shader and VFX Graph drive the simulation automatically.
- World-Space Drift: Particles spawn and float within the volume bounds using World Space simulation, keeping their natural drift relative to the environment regardless of camera movement.
- Demo Scene Integration: Open FloatingDust_ShowcaseScene to inspect how the dust volume behaves under typical lighting setups and camera movements.

5. Material & Particle Properties Reference

The visual and physical behavior of the dust can be fully customized directly from the Visual Effect component inspector via the exposed properties:

- **SpawnRadius**: Defines the radius of the spherical volume where dust particles are spawned.
- **Amount**: Controls the number of active particles spawned in the volume (from 0 up to the 2,000 maximum capacity).
- **Size**: Sets the maximum particle size variation across the system.
- **Alpha**: Adjusts the baseline opacity multiplier for all particles in the volume.
- **TurbulenceIntensity**: Controls the strength of the internal turbulence field for smooth drifting motion.
- **ColorGradient**: Random color and transparency gradient applied to particles upon spawning.